Activities 1 & 2: AI Models and Transformer Fundamentals

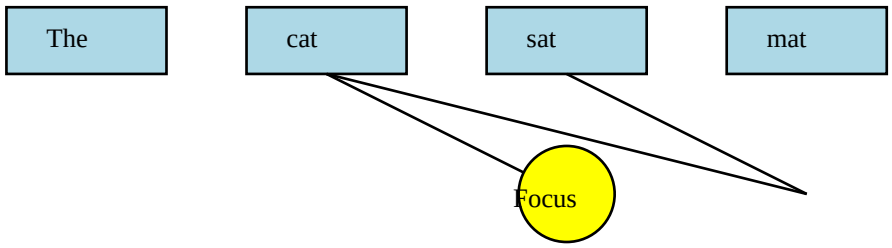
Activity 1 - Recent AI Models

Provider	Model	Release	Context Window	Use Case
OpenAI	GPT-5	2025	400K	Advanced reasoning & coding
Anthropic	Claude Opus 4	2025	200K	Complex workflows
Google	Gemini 2.5 Pro	2025	1M	Large document analysis
xAI	Grok 4	2025	256K	Research and reasoning
Meta	Llama 4 Maverick	2025	1M	Open-weight AI
Mistral	Magistral Small	2025	128K	Reasoning tasks
Cohere	Command A	2025	256K	Enterprise agents
AI21	Jamba 1.6 Large	2025	256K	Private deployment
Moonshot AI	Kimi K2	2025	262K	Agentic workflows
DeepSeek	DeepSeek V3.1	2025	164K	Low-cost reasoning

Activity 2 - Understanding Transformer Components

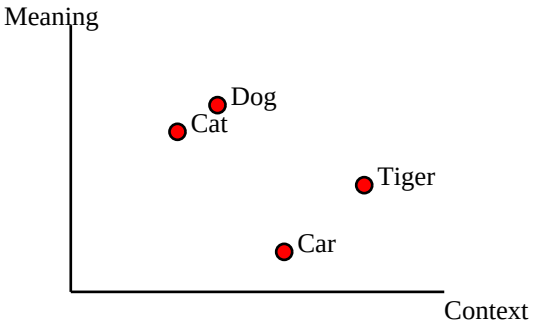
1. Self-Attention Mechanism

Self-attention allows a token to examine other tokens in the sentence and determine which are most important. It helps the model understand context and relationships between words.



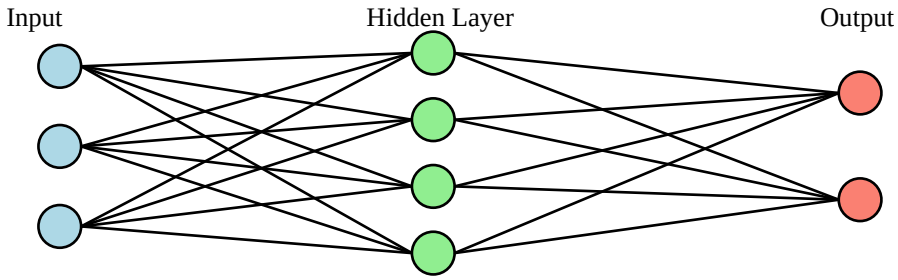
2. Vector Embeddings

Embeddings convert words into numerical vectors. Similar concepts are placed close together in vector space, allowing the model to understand meaning mathematically.



3. Feed Forward Neural Network (FFN)

After attention gathers information, the Feed Forward Neural Network processes it. The FFN applies mathematical transformations and learns patterns, facts, and relationships.



Memory Aid: Embeddings = Meaning, Self-Attention = Communication, FFN = Thinking.